

# RA-04: narrow-band stability and exact-tail deflation

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## Abstract

This continuation establishes a quantitative good-start graph for sufficiently narrow, nonzero-width leading spectral bands. Barycentric scaling eliminates dependence on the overall leading spectral condition number. A separate generic rank argument proves exact recovery when the spectrum below the target cutoff has few distinct values, with the boundary multiplicity handled explicitly. The unrestricted RA-04 bound is not established: the width restriction in the first theorem is substantive, and the second theorem does not cover an arbitrary diverse tail. The status is **PARTIAL**.

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## 1 Target and dependencies

Let  $M = AA^T$  have eigenvalues  $\lambda_1 \geq \dots \geq \lambda_n \geq 0$ . Write

$$t = \lceil k/b \rceil, \quad m = bt \leq \text{rank}(A), \quad \Delta = \min_{1 \leq i \leq m-b} \frac{\lambda_i - \lambda_{i+b}}{\lambda_i} > 0.$$

For  $m = b$ , the empty minimum is  $\Delta = 1$ . The Gaussian starting block is  $G \in \mathbb{R}^{n \times b}$ , and

$$\mathcal{K}_q(M, G) = \text{range}[G, MG, \dots, M^{q-1}G].$$

RA-04 requests, for the prescribed projection-and-SVD-truncation output, all three of its approximation guarantees with iteration order

$$q = O\left(\frac{t \log(2/\Delta) + \log(n/(\delta\varepsilon))}{\sqrt{\varepsilon}}\right). \quad (1)$$

The previous report obtained the all-input bound with  $t \log(2m/\Delta)$  in place of  $t \log(2/\Delta)$ , together with several exact target regimes.<sup>1</sup> The present work does not remove the additional  $t \log m$  for arbitrary spectra.

The new graph theorem below is self-contained. Its implication for all three RA-04 output guarantees uses the good-start convergence transfer of Chen et al., Definition 3.1, Imported Theorem 3.2 and Observation 3.3 of arXiv v2, exactly as imported in the previous report.<sup>2</sup> In particular, a graph  $U[I_m; F]$  at depth  $q_0$  with  $\log(1 + \|F\|_F^2) = O(t \log(2/\Delta) + \log(n/\delta))$  and  $q_0 = O(t)$  provides the good start used in that transfer. No independent new proof of the transfer theorem is claimed here. The fixed-spectrum corollary also uses the earlier all-input bound; this additional dependency is stated where used.

## 2 A Gaussian block estimate

**Lemma 2.1.** *Let  $H \in \mathbb{R}^{a \times b}$  be standard Gaussian,  $a \leq b$ . For  $u \geq 0$ ,*

$$\mathbb{P}\{\sigma_{\min}(H) \leq u\} \leq 16\sqrt{a}u.$$

*Proof.* Let  $v$  be a unit left singular vector for the smallest singular value, with an independent fair sign. Orthogonal invariance makes  $v$  uniform on the unit sphere and independent of the singular values. For a fixed coordinate,  $\mathbb{E}v_i^2 = 1/a$  and  $\mathbb{E}v_i^4 = 3/[a(a+2)]$ . Thus

$$\mathbb{P}\{|v_i| \geq (2a)^{-1/2}\} \geq \frac{1}{12}$$

by the Paley–Zygmund inequality. Repeated singular values have probability zero and do not affect this assertion.

Let  $D_i$  be the distance of row  $i$  from the span of the other rows. The Schur-complement identity gives

$$D_i^{-2} = ((HH^T)^{-1})_{ii} \geq \frac{|v_i|^2}{\sigma_{\min}(H)^2}.$$

Conditioned on the other rows,  $D_i$  has a chi distribution with  $b - a + 1$  degrees of freedom. Consequently  $\mathbb{P}\{D_i \leq z\} \leq \sqrt{2/\pi} z$ . Independence of  $v$  from the singular values now yields

$$\frac{1}{12} \mathbb{P}\{\sigma_{\min}(H) \leq u\} \leq \mathbb{P}\{D_i \leq u\sqrt{2a}\} \leq \frac{2}{\sqrt{\pi}} \sqrt{a}u.$$

Since  $24/\sqrt{\pi} < 16$ , the stated constant follows. □

<sup>1</sup>*RA04\_partial\_results.pdf*, supplied previous report; see also the source inventory at the end of this report.

<sup>2</sup>T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486; SODA 2026. <https://arxiv.org/abs/2508.06486>.

For  $r$  blocks with at most  $b$  rows each, a union bound shows that every least singular value is at least  $\eta/(16r\sqrt{b})$  except on an event of probability at most  $\eta$ . Independence between this event and a subsequent perturbation is not assumed or needed.

### 3 Deterministic barycentric scaling

Partition the first  $m$  indices into  $r \geq 2$  consecutive nonempty groups  $J_s$  with  $|J_s| \leq b$ . Let

$$c_1 > \cdots > c_r > 0, \quad \beta = \min_{s < r} \left(1 - \frac{c_{s+1}}{c_s}\right) > 0, \quad |\lambda_i - c_s| \leq wc_s \quad (i \in J_s).$$

Define the degree- $r-1$  polynomials and signed normalization constants

$$p_j(x) = \prod_{\ell \neq j} (x - c_\ell), \quad d_s = \prod_{\ell \neq s} (c_s - c_\ell).$$

**Lemma 3.1.** *If  $rw/\beta \leq 1/2$ , then, for  $i \in J_s$ ,*

$$\left\| \left( \frac{p_j(\lambda_i)}{d_s} \right)_{j=1}^r - e_s \right\|_2 \leq \frac{2rw}{\beta}. \quad (2)$$

Moreover,

$$\frac{\max_{j, 0 \leq x \leq (1+w)c_r} |p_j(x)|}{\min_s |d_s|} \leq \left( \frac{1+w}{\beta} \right)^{r-1}. \quad (3)$$

Neither estimate contains  $c_1/c_r$ .

*Proof.* For any distinct  $s, \ell$ ,  $|c_s - c_\ell| \geq \beta \max(c_s, c_\ell)$ . The diagonal ratio is

$$\frac{p_s(\lambda_i)}{d_s} = \prod_{\ell \neq s} \left( 1 + \frac{\lambda_i - c_s}{c_s - c_\ell} \right).$$

Each perturbation has absolute value at most  $w/\beta$ . Its difference from one is at most  $2(r-1)w/\beta$ . For  $j \neq s$ , isolate the factor  $(\lambda_i - c_s)/(c_s - c_j)$ ; the other factors have product bounded by  $\exp((r-2)w/\beta) \leq 2$ . The squared row error is therefore at most  $4[(r-1)^2 + (r-1)](w/\beta)^2 \leq 4r^2(w/\beta)^2$ .

For the tail estimate,  $|x - c_\ell| \leq (1+w)c_\ell$ , so

$$\max_j |p_j(x)| \leq (1+w)^{r-1} \prod_{\ell < r} c_\ell.$$

On the other hand, for every  $s$ ,

$$|d_s| \geq \beta^{r-1} \left( \prod_{\ell < s} c_\ell \right) c_s^{r-s} \geq \beta^{r-1} \prod_{\ell < r} c_\ell.$$

Division proves the claim. The cancellation is the reason to use this normalization rather than perturb a raw monomial Vandermonde matrix.  $\square$

## 4 A graph theorem for nonzero-width bands

**Theorem 4.1.** *Let  $\eta = \delta/3$ , where  $0 < \delta < 1$ . Under the preceding band assumptions, suppose*

$$w \leq \frac{\beta\eta^{3/2}}{64r^2b\sqrt{m}}. \quad (4)$$

*With probability at least  $1 - \delta$ ,  $\mathcal{K}_r(M, G)$  contains the columns of  $U[I_m; F]$ , where  $U$  is an eigenbasis of  $M$  and*

$$\|F\|_F \leq 32br^{3/2}\sqrt{n-m}\eta^{-3/2}\left(\frac{1+w}{\beta}\right)^{r-1}. \quad (5)$$

*The empty tail when  $n = m$  has norm zero.*

*Proof.* In the eigenbasis write  $U^T G = [H; T]$ . The polynomial Krylov matrix with basis  $p_1, \dots, p_r$  has head rows formed by concatenating  $p_j(\lambda_i)h_i^T$ . Divide each row in group  $s$  by  $d_s$  and call the result  $C$ . Its centered counterpart  $C_0$  is block diagonal with blocks  $H_s \in \mathbb{R}^{|J_s| \times b}$ . With  $E = C - C_0$ , Lemma 3.1 gives

$$\|E\|_2 \leq \|E\|_F \leq \frac{2rw}{\beta} \|H\|_F.$$

Intersect the events

$$\min_s \sigma_{\min}(H_s) \geq \frac{\eta}{16r\sqrt{b}}, \quad \|H\|_F \leq \sqrt{\frac{mb}{\eta}}, \quad \|T\|_F \leq \sqrt{\frac{(n-m)b}{\eta}}.$$

Lemma 2.1 and Markov's inequality bound total failure probability by  $3\eta = \delta$ ; omit the last event for an empty tail. On this intersection,  $R_0 = C_0^\dagger$  satisfies  $C_0 R_0 = I_m$  and  $\|R_0\|_2 \leq 16r\sqrt{b}/\eta$ . The width bound implies

$$\|ER_0\|_2 \leq \frac{32r^2b\sqrt{m}w}{\beta\eta^{3/2}} \leq \frac{1}{2}.$$

Thus

$$R = R_0(I_m + ER_0)^{-1}, \quad CR = I_m, \quad \|R\|_2 \leq \frac{32r\sqrt{b}}{\eta}.$$

This is a valid right inverse even for rectangular blocks.

Let  $D$  have diagonal entry  $d_s$  on  $J_s$ . The unscaled head is  $DC$ , so  $RD^{-1}$  is its right inverse. Multiplication of the full polynomial Krylov matrix by this matrix produces head  $I_m$  and tail  $F$ . The unscaled tail has Frobenius norm at most  $\sqrt{r} \max_{x,j} |p_j(x)| \|T\|_F$ . Hence

$$\|F\|_F \leq \sqrt{r} \|T\|_F \|R\|_2 \frac{\max_{x,j} |p_j(x)|}{\min_s |d_s|},$$

which is (5). Each basis polynomial has degree at most  $r - 1$ , so the constructed space lies in  $\mathcal{K}_r(M, G)$ . Although  $E$  and  $C_0$  use the same Gaussian rows, only deterministic inequalities on the stated event were used. There is no independence error.  $\square$

## 5 Consequences for the requested iteration order

Take  $r = t$  and  $|J_s| = b$ . Consecutive-band comparison gives

$$\beta - 2w \leq \Delta \leq \beta + 2w.$$

In particular,  $w \leq \beta/4$  implies  $\beta/2 \leq \Delta \leq 3\beta/2$ . Assumption (4) is stronger than this restriction. Since  $b, r, m \leq n$ , Theorem 4.1 yields

$$\log(1 + \|F\|_F^2) = O(t \log(2/\Delta) + \log(n/\delta)).$$

The imported good-start transfer therefore gives (1) with all three RA-04 guarantees. This proves the desired order for a nonzero-width regime, not for arbitrary widths. Unequal bands also give this consequence when  $r = O(t)$  and their center gap is comparable to  $\Delta$ ; the latter is then an additional hypothesis.

**Corollary 5.1.** *Suppose  $m \geq 2$ ,  $t \geq 2$ , there are  $t$  consecutive bands of size  $b$ , and*

$$w \leq \frac{\beta}{64 \cdot 3^{3/2} t^2 b m^{(3t+1)/2}}. \quad (6)$$

*Using the previous report's all-input bound, the RA-04 order (1) holds for every  $0 < \delta < 1$  and admissible  $\varepsilon$ .*

*Proof.* When  $\delta \geq m^{-t}$ , condition (6) implies (4). When  $\delta < m^{-t}$ , the extra  $t \log m$  in the previous all-input bound is at most  $\log(1/\delta)$  and is absorbed into the requested separate logarithmic term. These two cases cover all failure probabilities.  $\square$

The previous all-input theorem is an explicit dependency of this corollary, not a result re-proved in this continuation. No assertion is made that every admissible spectrum satisfies (6).

## 6 Generic block-Vandermonde rank

**Theorem 6.1.** *Let  $\rho$  rows be assigned real nodes, with at most  $b$  rows at each node. Let  $H \in \mathbb{R}^{\rho \times b}$  have a joint density. If  $qb \geq \rho$ , then*

$$[H, \Lambda H, \dots, \Lambda^{q-1} H]$$

*has row rank  $\rho$  almost surely.*

*Proof.* Assign each equal-node group to distinct currently least-loaded colors among  $b$  colors. The maximum and minimum loads differ by at most one after each group: increasing a subset of the minimum loads either preserves the two adjacent load levels, or raises all minima and some maxima to the next two levels. Consequently the final largest load is  $\lceil \rho/b \rceil \leq q$ .

For a witness, set each row of  $H$  equal to its color's coordinate vector. After permutations, the displayed matrix splits into ordinary Vandermonde blocks, one per color. Within a color the nodes are distinct and their number is at most  $q$ . Each block has full row rank, exhibiting a nonzero  $\rho \times \rho$  minor. That minor is a nonzero polynomial in the entries of  $H$ , whose zero set has Lebesgue measure zero.  $\square$

The coordinate-vector construction is a witness that a polynomial is not identically zero, not a claim about the Gaussian realization itself.

## 7 Exact recovery by distinct-tail deflation

**Theorem 7.1.** *Let  $\theta = \lambda_k > 0$ . Let  $a$  count eigenvalues strictly above  $\theta$ , and let  $d_\theta$  be the full multiplicity of  $\theta$ . Assume that every higher eigenvalue has multiplicity at most  $b$  and  $k - a \leq b$ . Put*

$$\rho = a + \min(d_\theta, b), \quad L = \#\{\text{distinct eigenvalues strictly below } \theta\}.$$

*Zero is counted once in  $L$  when present. Almost surely,  $\mathcal{K}_{L+\lceil \rho/b \rceil}(M, G)$  contains an optimal rank- $k$  left singular subspace. The prescribed projection-and-truncation algorithm then returns an exact best rank- $k$  approximation.*

*Proof.* Let  $f(x) = \prod_{\tau < \theta} (x - \tau)$ , with one factor for each distinct lower eigenvalue. The block  $f(M)G$  has zero components below  $\theta$  and nonzero deterministic row scalings in every higher eigenspace and at  $\theta$ . Each higher eigenspace has accessible dimension equal to its multiplicity almost surely. At  $\theta$  the accessible dimension is  $\min(d_\theta, b)$ .

Choose that many fixed coordinate rows from the  $\theta$  eigenspace and all rows from the higher eigenspaces. On these  $\rho$  coordinates the Gaussian block, after nonzero deterministic scalings, has a joint density. Each node has at most  $b$  chosen rows, so Theorem 6.1 gives full row rank after  $\lceil \rho/b \rceil$  blocks. If  $d_\theta > b$ , the remaining rows are linear combinations of the chosen  $b$  rows; since their eigenvalue is identical, this relation holds for every power. Thus the full filtered space has dimension exactly  $\rho$  and spans all accessible directions.

It contains every higher eigenspace and at least  $k - a$  directions at  $\theta$ , which is an optimal rank- $k$  left singular subspace. Multiplication by  $f$  adds  $L$  to the maximum polynomial degree, giving the asserted number of original Krylov blocks. Projection onto a space containing such a subspace, followed by the prescribed SVD truncation, is an exact best rank- $k$  approximation.  $\square$

RA-04's positive clustered gap implies the multiplicity assumptions: more than  $b$  copies above the cutoff, or more than  $b$  required copies at the cutoff, would give  $\lambda_i = \lambda_{i+b}$  within the first  $m$  entries. Also  $a \leq k - 1 \leq m - 1$ , so  $\rho \leq m + b - 1$ . Hence  $L + t + 1$  blocks suffice. This regime allows arbitrarily large lower-tail rank and multiplicities; it restricts the number of distinct lower values, not their total count.

### 7.1 A necessary boundary correction for exact recovery

Take  $b = 2$ ,  $k = m = 4$ , and eigenvalues  $(5, 4, 3, 2, 2, 2)$ . Then  $\Delta = 2/5$ ,  $L = 0$ , and  $\rho = 3 + 2 = 5$ . Three blocks suffice by Theorem 7.1; two generically do not. In  $\mathcal{K}_2$ , the block at eigenvalue 2 is  $H_{\text{tail}}(u + 2v)$ . Since  $H_{\text{tail}}$  has rank two almost surely, a vector supported only above the cutoff must have  $u + 2v = 0$ . The intersection with that higher eigenspace therefore has dimension at most two. Exact rank-four approximation would require all three directions strictly above 2, which is impossible in this intersection. This is not a counterexample to the big- $O$  assertion in RA-04.

## 8 Validation and remaining obligations

The exact checker uses rational arithmetic to test balanced-coloring rank witnesses, polynomial normalization, right-inverse identities, nonzero-width graph estimates, and the boundary example.

The optional NumPy checker supplies floating-point diagnostics. Execution results are recorded in the archive's `results/` directory; no test count is assumed in this prose.

These finite checks do not replace the proofs or independently establish the imported convergence theorem. They do not prove the unrestricted interpolation estimate or certify a universal iteration constant. The width assumptions remain essential to the presented argument. The prior general bound still contains an extra  $t \log m$  outside the proved regimes.

The sufficient square-depth interpolation estimate in the previous report is still unproved here. Additional Krylov powers might bypass it; even a counterexample to that stronger estimate would not by itself refute RA-04. The appropriate status of this archive is **PARTIAL**, not SOLVED.

## Sources

1. Open Problems in Numerical Linear Algebra, RA-04 repository entry. <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/randomized-and-low-rank-approximation/RA-04>.
2. *RA04\_partial\_results.pdf*, previous report supplied in this conversation. Used for the earlier all-input bound and the convergence-transfer dependency. See the immutable prior submission and its independent audit at prior submission (immutable revision e15ce48).
3. T. Chen, E. N. Epperly, R. A. Meyer, C. Musco, and A. Rao, *Does block size matter in randomized block Krylov low-rank approximation?*, arXiv:2508.06486; SODA 2026. <https://arxiv.org/abs/2508.06486>.
4. R. A. Meyer, C. Musco, and C. Musco, *On the Unreasonable Effectiveness of Single Vector Krylov Methods for Low-Rank Approximation*, arXiv:2305.02535. <https://arxiv.org/abs/2305.02535>.
5. N. Shao, *A structural bound for cluster robustness of randomized small-block Lanczos*. arXiv:2507.10144. <https://arxiv.org/abs/2507.10144>. Related work, not a proof of the unresolved estimate.